

Linear Algebra Concepts with Solved Numerical Problems

1. Norms and Distance

Example 1:

Find the norm of vector $v = (6, 8)$

Solution:

$$\|v\| = \sqrt{6^2 + 8^2}$$

$$\|v\| = \sqrt{36 + 64}$$

$$\|v\| = \sqrt{100}$$

$$\|v\| = 10$$

Example 2:

Find the distance between $u = (1, 2)$ and $v = (4, 6)$

Solution:

$$u - v = (-3, -4)$$

$$\text{Distance} = \sqrt{9 + 16} = 5$$

Simple Diagram:

$$v(4, 6)$$

*

|

| Distance = 5

|

*-----

u(1,2)

2. Dot Product and Orthogonality

Example:

$u = (2,3)$

$v = (3,-2)$

$$u \cdot v = (2)(3) + (3)(-2)$$

$$u \cdot v = 6 - 6 = 0$$

Since dot product = 0,

Vectors are orthogonal.

3. Eigenvalues and Eigenvectors

Example:

$$A = \begin{bmatrix} 4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 5 \end{bmatrix}$$

Step 1: Characteristic equation

$$\det(A - \lambda I) = 0$$

$$(4 - \lambda)(5 - \lambda) = 0$$

Eigenvalues:

$$\lambda_1 = 4$$

$$\lambda_2 = 5$$

Step 2: Eigenvectors

For $\lambda = 4$:

$$(4 - 4)x = 0$$

$$\text{Eigenvector} = (1, 0)$$

For $\lambda = 5$:

$$(5 - 5)y = 0$$

$$\text{Eigenvector} = (0, 1)$$

4. Orthogonal Matrix Example

$$Q = \begin{bmatrix} 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$Q^T = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 \end{bmatrix}$$

$$Q^T Q = I$$

Hence Q is orthogonal.

This represents 90 degree rotation.

Simple Rotation Diagram:

Before Rotation: After Rotation:

$$(1,0) \text{ ----> } (0,1)$$

5. Diagonalization

Example:

$$A = \begin{bmatrix} 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 \end{bmatrix}$$

Since eigenvalues are 2 and 3,

$$D = \begin{bmatrix} 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 \end{bmatrix}$$

P = Identity matrix

Now compute A^3 :

$$D^3 = \begin{bmatrix} 8 & 0 \\ 0 & 27 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 27 \end{bmatrix}$$

Thus,

$$A^3 = \begin{bmatrix} 8 & 0 \\ 0 & 27 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 27 \end{bmatrix}$$

Diagonalization makes power calculation easy.

Additional Concept Check Problems

1) Find norm of (3,4,12)

Answer: 13

2) Are (1,1) and (1,-1) orthogonal?

Answer: Yes

3) If $\lambda = 2$ is eigenvalue of matrix A, what does it mean?

Answer: Matrix scales eigenvector by factor 2.